The Principle of Shared Space

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2026

Axiom

$$c^2 = c + 1$$

The unique positive solution is $\varphi = (1+\sqrt{5})/2$, giving contraction rate $r = 1/(2\varphi) \approx 0.3090$.

1 Introduction

Shared space is the foundation upon which the very fabric of reality is built. Spacetime emerges from the infinite possibility of light updating topological information, contorted by the probability of matter.

$$\begin{aligned} \text{Light:} \quad & (1-r)^2 && = 0.4775 \\ \text{Matter:} \quad & r^2 && = 0.0955 \end{aligned}$$

This contortion produces the boundary channel — the structural boundary within which our visible universe resides.

$$\text{Boundary:} \quad 2r(1-r) = 0.4271$$

A forced relationship of simplicity along one initial axis: the axis of acknowledgement.

2 The Axis of Acknowledgement

Acknowledgement is the core of Unified Mechanics. Not just in terms of what acknowledgement is, but how, over time, even the simplest recognition of two states can give rise to complex relationships. Even the most naive or ignorant forms of acknowledgement can generate intricate boundaries and deeply layered structures.

The early universe is the clearest expression of this principle in action. The cosmic soup of that time was a hot, dense field of overcomplex geometry — light and matter so intertwined they were effectively indistinguishable. No distinct relationships had yet formed. But as expansion progressed, the uncoupled, overcomplex geometry of matter stretched and cooled. Through this process, light was able to saturate the geometry and finally map the probability of the substrate.

3 The Axis of Exploration

Spacetime, as a three-dimensional structure, allows full spherical freedom of traversal from any origin — complete freedom to expand along whatever path least action deemed most viable for the intended direction of its boundaries. This gave rise to distinct temperature and pressure gradients, continuously complexifying the boundaries until, eventually, you arrive at the complexity of the human being.

Just like humans who spend entire lifetimes building specialisations toward highly specific fields, you are free to explore the full extent of your parameter space. Moving into a new space, however, demands adaptation at the very least, and in more extreme cases, complete restructuring. Life presents a myriad of possibilities, each more complex than the last. Yet one unalienable truth remains: all human problems are inherently human. They always carry the shape of human society within them. We simply do not know any different.

4 The Axis of Familiarity

A hydrogen atom will not spontaneously become moscovium under normal conditions. Moscovium can, however, be synthesised under precise laboratory constraints. In the same way, a gorilla can be made comfortable within a zoo enclosure — the familiar environment carefully simulated to meet its needs. The conditions of both the entity and its native environment must be satisfied for stability to emerge.

This is the final complex relationship the universe had to traverse once probability had solidified its influence on existence, narrowing down what could be into what will be.

And what remains is the intricate reflection of what is.

From individual grains of sand on a beach to exoplanets at the edge of perception, we all share in this intricate relationship within a place we simply call space.

The three axes of this paper map directly onto the three channels of the $c^2 = c + 1$ recursion:

Axis	**Channel**	**Weight**
Acknowledgement	Boundary	$2r(1-r) = 0.4271$
Exploration	Light	$(1-r)^2 = 0.4775$
Familiarity	Matter	$r^2 = 0.0955$

The boundary channel and these three axes compose reality. The intricate web of information that occupies spacetime.